

Question Paper

Exam Date & Time: 31-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Operating Systems [IT_3128]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply your understanding of *multiprogramming* and *timesharing* to explain how CPU utilization and system responsiveness differ between the two. (4)

- A) Construct an example scenario where timesharing provides a clear performance advantage over multiprogramming.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Apply your understanding of process states and schedulers to explain how a process transitions among different states. (4)

Develop a scenario where multiple processes share the CPU, and illustrate how different schedulers coordinate to maintain efficient CPU utilization.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) The following C code snippet is given: (2)

```
#include <stdio.h>
#include <unistd.h>
```

```
int main() {
    pid_t pid = fork();
    if (pid == 0)
        printf("Child process endsn");
    else {
        printf("Parent runningn");
        sleep(10);
    }
    return 0;
}
```

Identify the state of the child process after termination and before the parent process exits.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Four processes arrive at time 0 with the following burst times: (4)
P1 = 5 ms, P2 = 10 ms, P3 = 15 ms, P4 = 20 ms.

- A) Assume the time quantum for Round Robin is 5 ms, and context switch time is negligible. Compare the average waiting time for these processes under Preemptive SJF and Round Robin scheduling. Analyze which algorithm gives a better trade-off between CPU utilization and response time, providing a short justification.

- B) Two processes P0 and P1 are implementing Peterson's solution. (4)
 At some point, the variables hold the following values:
 turn = 0;
 flag[0] = true;
 flag[1] = true;
 and both processes are in their while (flag[j] && turn == j) loop.
 Construct a sequence of events to show which process will enter the critical section first, and justify why the three critical-section requirements are or are not violated under this interleaving.

- C) Compare the function of binary and counting semaphores in terms of the number of resources managed and their effect on process synchronization. (2)
 Provide one example scenario where using a counting semaphore is essential over a binary semaphore.

- 3) A system has four processes (P1–P4) and two resource types (R1, R2). (4)

- A) The total instances of R1 = 3 and R2 = 3.
 The current Allocation and Request matrices are:

Process	Allocation (R1,R2)	Request (R1,R2)
P1	1, 0	0, 1
P2	1, 1	1, 0
P3	1, 2	0, 0
P4	0, 0	1, 1

- Develop a suitable model of the system using the Deadlock Detection Algorithm.
- Apply the algorithm to select which processes can finish and which cannot.

Determine whether a deadlock exists, and organize your findings with justification.

- B) A system uses the Banker's Algorithm for resource allocation. Process P1 currently holds some resources and makes a new request for additional resources. The operating system must check whether granting this request keeps the system in a safe state. (4)
 Construct the sequence of logical steps the operating system should follow to solve this problem using the Banker's safety algorithm.
 Explain how the system can utilize the "pretend allocation" step to identify if the new request should be granted or deferred.

- C) A developer is analyzing OS activities such as: (2)
1. Allocating memory for new processes,
 2. Managing access permissions for files,
 3. Scheduling disk operations, and
 4. Executing I/O requests.
- Classify each of the above tasks under the appropriate function of the operating system.

- 4) A system has a memory of 300 KB divided into the following free blocks (holes): (4)
 Hole 1 = 70 KB, Hole 2 = 120 KB, Hole 3 = 40 KB, and Hole 4 = 60 KB.
 A) Four processes arrive sequentially with memory requirements:
 P1 = 50 KB, P2 = 80 KB, P3 = 30 KB, P4 = 60 KB.
 Apply Best-Fit and Worst-Fit algorithms to allocate memory to these processes.
 Show the allocation table and final memory state after all allocations.
 Finally, identify which strategy results in less total external fragmentation.

- B) A computer system uses base and limit registers for memory protection and relocation. (4)
- The base register of a process contains the value 40,000.
 - The limit register contains the value 12,000.
 - A user process generates the following logical addresses: 10,500, 14,500, and 12,200.

Develop the address translation scheme for this system and determine which of the above addresses will cause a trap to the operating system.
 Further, explain how the MMU (Memory Management Unit) utilizes these registers during context switching to ensure memory protection.

- C) Two processes, P₁ and P₂, share the same physical memory space using a global page replacement algorithm. (2)
 During execution, you observe that P₁'s page faults increase dramatically whenever P₂ starts execution.
 Contrast the behavior of global and local page replacement strategies in this scenario.
 Justify with proper reasons which approach would better control thrashing.

- 5) A process with 4 page frames executes the reference string: (4)
 0, 1, 2, 3, 2, 1, 4, 0, 1, 2, 3, 4.
 A) All frames are initially empty.
 Contrast the number of page faults obtained using the FIFO and Second-Chance (Clock) algorithms.
 Show the intermediate state of the reference bit (use binary notation), and conclude whether Second-Chance successfully reduces unnecessary replacements.

- B) A disk has 200 cylinders (0-199). The disk arm is currently at cylinder 100, moving toward the higher-numbered cylinders. (4)
 Pending I/O requests are for cylinders:
 176, 79, 34, 60, 92, 11, 114, 139, 150.
 Construct the sequence of servicing requests using both SCAN and C-SCAN scheduling algorithms.
 Compute the total head movement in each case and identify which algorithm provides more uniform waiting time.

- C) Apply your understanding of file access methods to identify which method would be most efficient for: (2)
 a) A compiler accessing variable symbol tables.
 b) A media player reading large continuous audio files.
 Briefly justify your choice in each case.

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

IT_3128:Operating Systems

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	18
3	3	CO 3	18
4	4	CO 4	2
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

IT_3128:Operating Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	36(72%)
4	Analysis	5	14(28%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)